

Hack for MS zune HD released

The first hack ever for Microsoft Zune HD was released. Named Open ZDK, it allows full access to the Zune hardware features

Camera RAW 6.1 released

Adobe releases RC version of its Camera RAW 6.1 raw processing software with lens correction feature with profile-based corrections

waves. The human anatomy is shown in black and white whereas the blood flow is colour-coded. This hand-held device is great for a fast inspection of the urinary bladder, heart and abdominal organs. And it does all this in real time. So no more waiting at the doctor's clinic with a bursting bladder before the ultrasound scanning.

Let the brain do the communication

Moving from portable health monitoring to portable wearable brain signal mapper. A team, led by University of Maryland's (UM) associate professor of kinesiology, Dr. Jose Contreras - Vidal, have successfully reconstructed 3D hand motion from the brain signals. What's so great about it one may think, but there's a catch. The brain signals have been captured in a non invasive manner. Using a process called electroencephalography (EEG), which measures brain signals with the help of electrodes attached to the scalp.

The UM team placed an array of 34 sensors on the scalps of five volunteers to record their brain's electrical activity. The volunteers were asked to reach from a centre button to eight other buttons randomly around 10 times and the hand motions and brain signals were recorded. EEG recordings were analysed and correlated with the readings from motion sensors attached to the volunteers hands to create a 3D model of the hand movements. This has proved the maxim, that EEG signals are not strong enough to lead to any movement of limbs, wrong.

According to Dr. Contreras-Vidal, the research is aimed at understanding, how the multiple models of interaction between the brain and the body are developed and how these models are affected by diseases.

"We are currently applying our decoding methods to bipedal locomotion. We believe it is possible to use noninvasive brain signal (derived from EEG) to control lower limb exoskeleton to restore gait," said Dr. Vidal.

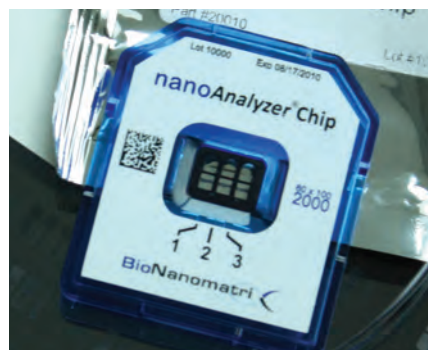
This research is a boon for patients who have lost the use of their limbs due to accident, they can control their prosthetics and equipments such as wheelchair with this. If imagination is stretched a wee bit more, this headset can also be used to give input

commands to a computer in a handsfree manner just by thinking.

As for the availability of the product in the market, Trent Bradberry, a research student working in Dr Vidal's team said, "There are three main steps that need to be done before our research enters the mainstream healthcare segment. First of all we need to create a near real-time implementation of our decoder within a brain-computer interface system, which is nearly complete. Secondly, test our system with persons that have an amputation or a spinal cord injury. Finally, transform our laboratory prototype to a polished product that is easy to use in the home." Dr. Vidal expects this technology to be translated in some portions of the healthcare world within the next five years.

Genome Sequencing in \$100?

Researchers at the BioNanomatrix, Pennsylvania under the leadership of Mr. Han Cao are working on a nanofluidic chip which claims to bring down the Genome sequencing cost to just \$100. A working prototype is ready and the technology has just been patented in March 2010. The nanoanalyser set up comprises three parts viz : the nanoanalyser chip, the software to read the data and an imaging device to capture it on a CCD sensor. Since BioNanomatrix only unravels the DNA, its working in collaboration with Complete



Genomics, a Silicon Valley based company, which handles the sequencing aspect.

It works like this - a large coiled chromosomal DNA strand is inserted in the fluid. The whole area is divided into patterned bio chips that help the coils to untangle such that when the strands reach the final nanochannel array, they are separated from each other and each array (100 nm wide)

can only house a single strand. These gaps are at the micrometer level in the beginning and gradually reach the nanometer level. The fluid guides the strands.

Unlike other genome sequencing processes in which the strands are cut and then re stitched to be analysed, the nano-analyser does not break the strand at any point. "Human genome has recently been discovered to have highly complex genomic sequence structural variations. The location, the orientation of specific genes or blocks of genes are different among individuals. These Structural Variations (SVs) have been linked to many complex trait diseases such as autoimmune, autism, cancer etc. Frag-



"The method we have implemented, enables one to keep the medically relevant contextual structural information intact by preserving the direct imaging long strands of DNA at a single molecule level without any amplification." Mr Han Cao, Founder and Chief Scientific Officer, BioNanomatrix

mented small reads leads to loss of information and their reassembly is expensive," said Dr. Cao stressing on the need to analyse strands without breaking them.

The initial applications of this research involve DNA sample integrity, DNA damage profiling and analysing the structural variations in the genome. "More accurate information with less cost enabled by this kind of new generation technology will make genomic analysis a routine method in clinical genetic diagnosis, prognosis and therapeutic development. It will also make personalised medicine finally possible," said Mr Cao. This research is funded by a five year \$8.9 million NIST - Advanced Technology Program (ATP) grant. As Complete Genomics Inc. (CGI) is already delivering low cost genome sequencing, BioNanomatrix expects the price to come down to the \$100 range in the next five years. **d**